Muhammad Irfan Fahmi

Computer Engineering | AI, Software & Embedded Systems
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Professional Summary

Computer Engineering background at UTeM, connecting software, intelligent systems and hardware through practical projects. Hands-on work spans computer-vision checkout validation, Android release engineering and Arduino-based sensing, supported by technical troubleshooting experience.

Education

Universiti Teknikal Malaysia Melaka (UTeM), Bachelor of Computer Engineering with Honours — Oct 2022 – Nov 2026

- Graduation scheduled: 2 November 2026.

Politeknik Port Dickson, Diploma in Electronic Engineering (Computer) — Dec 2018 – May 2022

Kolej Komuniti Selandar, Certificate in Computer Systems and Networking — Jul 2017 – Feb 2019

Technical Skills

Programming & data: Python, C/C++, JavaScript, TypeScript, SQL, MySQL, Git.

Intelligent systems: YOLOv8, OpenCV, MediaPipe, Random Forest, LLM orchestration.

Hardware & systems: Arduino, HX711, load cells, sensor integration, Android investigation, Windows troubleshooting, VLAN, OSPF, Wireshark.

Selected Projects

Hybrid Self-Checkout

- Built a Python pipeline combining YOLOv8/OpenCV product detection, USB barcode input and MySQL product/transaction data.
- Repository-reported evaluation: 77.4% precision and 72.0% recall after 50 epochs; tested visual verification challenges including occlusion and similar packaging.

SMC Crows / Android Release Engineering

- Worked on Android compatibility investigation, release packaging, device testing and a community distribution portal.
- Documented installation and account backup/recovery workflows; release-specific test records explain device and Android-version constraints.

IoT Livestock Weight Tracking

- Contributed to a team weighing/monitoring platform using Arduino, HX711 load-cell amplifiers, C/C++ signal filtering and telemetry dashboards.
- Team awarded Third Place at INOTEK 2025 (Series 1), 15 January 2025, for Development of Livestock Weight Tracking System Based on IoT.

Technical Experience

Technical Staff, Global Elite Ventures Sdn. Bhd — Sep 2021 – Jul 2022

Technical troubleshooting and hardware/software deployment.

Contract Assistant Engineer, ARNN Technologies & Karar Solution — Jun 2020 – Jul 2020

Engineering project support and technical problem-solving.

Technician, OKCS Seri Kembangan — Jun 2018 – Nov 2018

Diagnostics, repairs, hardware maintenance and software configuration.

Professional Registrations

BEM: Graduate Engineer, Computer branch; approved 15 October 2026. Public directory.

MBOT: Graduate Technologist, Information & Communication Technology (IT); GT26100872; approved 22 October 2026.

Selected Training

Festo / STDC: Industrial Automation with AI in Manufacturing, 14-16 July 2026.

Cisco Networking Academy: Enterprise Networking, Security, and Automation; Switching, Routing, and Wireless Essentials (23 February 2026). Course completions.

Languages

Bahasa Malaysia (native); English (professional working proficiency).